

VERTICAL JUMP BIOMECHANICS AFTER PLYOMETRIC, WEIGHT LIFTING, AND COMBINED (WEIGHT LIFTING + PLYOMETRIC) TRAINING

FOTINI ARABATZI,¹ ELEFATHERIOS KELLIS,¹ AND EDUARDO SAÈZ-SAEZ DE VILLARREAL²

¹Laboratory of Neuromechanics, Department of Physical Education and Sport Science, Aristotle University of Thessaloniki at Serres, Serres, Greece; and ²Laboratory of Human Performance, Faculty of Sport, University Pablo de Olavide, Seville, Spain

ABSTRACT

Arabatzi, F, Kellis, E, and Saèz-Saez de Villarreal, E. Vertical jump biomechanics after plyometric, weight lifting, and combined (weight lifting + plyometric) training. *J Strength Cond Res* 24(9): 2440–2448, 2010—The purpose of this study was to compare the effects of an Olympic weight lifting (OL), a plyometric (PL), and combined weight lifting + plyometric (WP) training program on vertical jump (VJ) biomechanics. Thirty-six men were assigned randomly to 4 groups: PL group ($n = 9$), OL group ($n = 9$), WP group (10), and control (C) group ($n = 8$). The experimental groups trained 3 d·wk⁻¹, for 8 weeks. Sagittal kinematics, VJ height, power, and electromyographic (EMG) activity from rectus femoris (RF) and medial gastrocnemius (GAS) were collected during squat jumping and countermovement jumping (CMJ) before and after training. The results showed that all experimental groups improved VJ height ($p < 0.05$). The OL training improved power and muscle activation during the concentric phase of the CMJ while the subjects used a technique with wider hip and knee angles after training ($p < 0.05$). The PL group subjects did not change their CMJ technique although there was an increase in RF activation and a decrease of GAS activity after training ($p < 0.05$). The WP group displayed a decline in maximal hip angle and a lower activation during the CMJ after training ($p < 0.05$). These results indicate that all training programs are adequate for improving VJ performance. However, the mechanisms for these improvements differ between the 3 training protocols. Olympic weight lifting training might be more appropriate to achieve changes in VJ performance and power in the precompetition period of the training season. Emphasis on the PL exercises should be given when the competition period approaches, whereas the combination of OL and PL exercises

may be used in the transition phases from precompetition to the competition period.

KEY WORDS power, EMG, training adaptations, performance, jumping

INTRODUCTION

Vertical jumping (VJ) performance is determined by a complex interaction among several factors including maximal force capacity, rate of force development, muscle coordination, and stretch-shortening cycle (SSC) use (25). Various training methods have been applied to improve VJ performance. Such methods include plyometric (PL) training, resistance training, weight lifting (WL) training, and electrical stimulation (16,18,31,32). Several studies have attempted to identify the mechanisms responsible for improvement in VJ performance (26,27). Among various mechanisms, specificity of training is considered as an important criterion to determine optimal training strategies for improvements in jump performance (27).

Plyometric training refers to the performance of SSC movements that involve a high-intensity eccentric contraction immediately before a rapid and powerful contraction (19). The effects of PL on VJ performance have been extensively studied. Particularly, some studies have demonstrated improvements in VJ height after PL exercise (1,21,28,31,32), whereas others have reported nonsignificant effects (8,12,35). A recent meta-analysis has shown that the effects of PL on performance are higher for slow SSC jumps but lower for fast SSC actions or squat (concentric) jumps (SJ) (21). Although the mechanisms by which PL improves VJ are not fully understood, it has been suggested that PL increases jumping height because of a greater muscle activation level and a higher mechanical output of the contractile machinery (1,16). Particularly, the power produced when performing PL movements enables the hip, the knee, and the ankle joints to reach high angular velocities at the end of the push-off phase (2). Previous studies (1,32) have shown that training with loaded jump squats resulted in higher VJ height improvements compared with combined weight and drop-jump training. This combination develops more power, which is analogous to the velocity of the contraction (5).

Address correspondence to Fotini Arabatzi, farabaji@phed-sr.auth.gr.
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Several investigations indicated that the movement pattern of Olympic weight lifting (OL) movements (snatch, clean, and jerk) is similar to that of a VJ (9,10,16). Particularly, lifters have been shown to have a higher VJ height and power output compared to other athletes (16). For this reason, OL movements have been considered as effective for improving VJ ability (9,10). Olympic weight lifting exercises may be more specific to the VJ than traditional weight training exercises. However, Hoffman et al. (13) found nonsignificant differences between WL and traditional strength training effects on power and speed development. Therefore, there is not enough evidence to conclude that OL training improves VJ height.

It still remains unclear how the OL overload could affect the velocity of the movement, and the spatiotemporal characteristics of neuromuscular coordination during VJ. It seems reasonable to hypothesize that OL is associated with certain aspects of VJ performance. To our knowledge, there is only 1 study that compared the effects of PL and WL training on VJ performance (30). Particularly, Tricoli et al. (28) found that OL training significantly improved sprint speed, SJ, countermovement jump (CMJ), and half-squat 1 repetition maximum (1RM) performance, whereas PL training improved only CMJ and half-squat 1RM efforts. Therefore, it appears that the short-term effects of OL training seem to be more beneficial for improvement in jump performance than traditional jump training.

Because of the multifaceted nature of VJ performance, a single training method approach may not be as effective as combining training methods to provide variation in stimulus and to increase the overall training adaptation (16). Many studies have found that the combination of traditional weight training and PL training may provide a greater stimulus to VJ performance (1,27). In a recent study, however, Mangine et al. (20) have shown that the addition of ballistic exercises to traditional heavy resistance training augmented maximal strength but not jump squat power. Adaptations to VJ appear specific to the movement pattern and velocity of the exercises used in training (31). To our knowledge, the effects of combined weight lifting + plyometric training (WP) on muscle activation patterns, ground reaction forces (GRFs), and kinematics during VJs have not been thoroughly investigated. Investigating the effect of each type of exercise program and their combination may provide additional useful information on the type of adaptations that occur after each training program. In turn, this will allow the coach to decide when to use each type of exercises within the year plan of the training program (long-term periodization). The purpose of this study was to compare the effects of 3 training programs on VJ biomechanics: a PL program, an OL program, and a combined PL and OL program.

METHODS

Experimental Approach to the Problem

The study was designed to compare training adaptations in jump performance and electromyographic (EMG) activity between 3 training programs. To investigate this question, we matched healthy students of physical education and subsequently randomly divided them into 3 training groups: (a) a WL group, (b) a PL group, (c) a combined WP group. This design enabled us to examine whether or not the addition of WL training to PL training augmented jump performance to a greater extent than WL and PL training alone. The outcome variables were VJ height, mechanical power, kinematics, and EMG activity of rectus femoris (RF) and gastrocnemius (GAS).

Subjects

This study involved a total of 36 hale male students of physical education (age = 20.3 ± 2 years, height = 184.8 ± 8.3 cm, and mass = 85.2 ± 6.8 kg) all free from any musculoskeletal or neurological disease or impairment. They were allocated randomly to 1 of 4 groups and did not differ significantly ($p > 0.05$) in any of the dependent variables. All subjects were physically highly active and experienced in performing various jumps through participation in various sport activities through their regular academic program. All subjects had at least 1 year of experience in resistance training, but they did not systematically perform strength drills and jumps. The local University Ethics Committee approved the study. All subjects were informed of any risk and signed an informed consent form, before their participation.

Study Design

This was a longitudinal study involving 4 groups (Table 1). The first group performed OL exercises. The second group performed PL drills, and the third group performed both OL exercises and PL drills. The last group served as the control (C) group. All groups were tested before and after an 8-week training period (February–March). The training groups attended 24 sessions (3 sessions per week) in total. Before the testing period, subjects of the OL, WP, and PL groups underwent 6 familiarization sessions (3 training sessions for familiarization with movement technique and 3 familiarization sessions with the experimental protocol). The controls performed the laboratory familiarization only.

Training Protocol

A standardized warm-up including jogging, stretching exercises, and several WL repetitions and jumps was provided for all experimental groups before the beginning of each training session.

Weight Lifting Program. The training program consisted of 5 Olympic-style WL exercises: snatch from a squat position, high-pull, power clean, half-squat, and clean and jerk. During the first 2 weeks, the subjects performed 4 sets of 4 repetitions of each exercise. The initial load corresponded to 75% of 1RM, and 3 minutes of rest was provided between sets.

TABLE 1. Exercises and volume of training for each experimental group.*

Groups	Exercises	Weeks 1–4	Weeks 5–8
Weight lifting	Power clean	4–6 × 6RM	4 × 4RM
	Snatch	4–6 × 6RM	4 × 4RM
	Clean and jerk	4–6 × 6RM	4 × 4RM
	High pull	4–6 × 6RM	4 × 4RM
	Half-squat	4–6 × 6RM	4 × 4RM
Plyometrics	Double-leg hurdle hops	4–6 × 6R	6 × 6R
	Alternated single-leg hurdle hops	4–6 × 6R	6 × 6R
	Double-leg hops	4–6 × 6R	6 × 6R
	Half-squat	4–6 × 6RM	4 × 4RM
Combined	Power clean (2 × Week)	4–6 × 6RM	4 × 4RM
	Snatch (2 × Week)	4–6 × 6RM	4 × 4RM
	Clean and jerk (2 × Week)	4–6 × 6RM	4 × 4RM
	High pull (2 × Week)	4–6 × 6RM	4 × 4RM
	Half-squat (2 × Week)	4–6 × 6RM	4 × 4RM
	Double-leg hurdle hops (2 × Week)	4–6 × 6R	6 × 6R
	Alternated single-leg hurdle hops (2 × Week)	4–6 × 6R	6 × 6R
	Double-leg hops (2 × Week)	4–6 × 6R	6 × 6R
		4–6 × 6RM	4 × 4RM

*6RM = 6 repetition maximum strength; R = repetitions.

For weeks 3 and 4, the volume of training significantly increased to 4 sets of 6 reps at 80% of 1RM. For the last 4 weeks, 4 sets of 4 reps at 80–90% of 1RM were performed.

Plyometric Training Program. The training program consisted of 4 PL drills: double-leg hurdle hops, alternate single-leg hurdle hops, double-leg hops, and half-squats. During the first 2 weeks, the PL group performed 4 sets of 6 reps of each exercise with the arms freely hanging on the side. The hurdle height used for the double-leg hurdle hops was 68 cm. For weeks 3 and 4, 6 sets of 6 reps were performed with an increase of hurdle height to 74 cm. Finally, for the last 4 weeks, the performance speed was increased by reducing the distance between hurdles (from 3.15 to 2.70 m). A standardized warm-up including jogging, stretching exercises, and several WL repetitions and jumps was provided for both experimental groups before beginning of each training session.

Weight Lifting + Plyometric Training Program. The training program consisted of 5 Olympic-style WL exercises: snatch from a squat position, high-pull, power clean, half-squat, clean and jerk, and 4 PL drills: double-leg hurdle hops, alternate single-leg hurdle hops, double-leg hops, and half-squats. During the first 2 weeks, the WP group performed 4 sets of 6 reps of each exercise. For WL exercises, the initial load corresponded to 75% of 1RM, and 3 minutes of rest was provided between sets. For weeks 3 and 4, the volume of training significantly increased to 4 sets of 6 reps at 80% of 1RM. For the last 4 weeks, 4 sets of 4 reps at 80–90% of 1RM were performed. For PL exercises, the hurdle height used for the double-leg hurdle hops was

68 cm. For weeks 3 and 4, 6 sets of 6 reps were performed from a 74-cm hurdle height. Finally, for the last 4 weeks, the intensity increased by reducing the distance between hurdles (from 3.15 to 2.70 m).

Instrumentation

Ground Reaction Forces. All VJs were performed on a Kistler piezoelectric force platform (Type 9281C, Kistler Instruments, Winterthur, Switzerland). The force platform was interfaced through Kistler amplifying units (Type 233A) to an Ariel Performance System (3), and GRFs were A/D converted at a sampling rate of 1,000 Hz.

Electromyography. Bipolar surface electrodes (Motion Control, IOMED Inc., voltage range: ± 4 – ± 12 V) in the form of metallic bars with a 1-cm interelectrode distance, interfaced to a 16-channel analog amplifier (sampling frequency 1,000 Hz, CMRR 100 db at 50/60 Hz, bandwidth 8,500 Hz, gain 400), were used to record the EMG activity of RF and medial GAS muscles. Electrode locations were prepared by shaving the skin of each site and cleaning it with alcohol wipes. Electrode placement was carefully measured and marked with permanent nonremovable ink to exactly ensure the same position before and after training.

Kinematics and Filters. For motion analysis, 2 video cameras (Panasonic AGI88 Tokyo, Japan, frame rate 60 Hz with a high-speed shutter) and a video recorder (Panasonic S500) were used. The cameras were placed at a 90° angle at a focal distance of 8 m. Skin markers were placed at 5 locations on the body: trunk (midaxillar line at umbilicus height), hip (superior

part or greater trochanter), knee (lateral epicondyle), ankle (lateral malleolus), and foot (head of the fifth metatarsal). The video image of a calibration frame was recorded before each measurement, and 8 calibration points were digitized to determine the 3-dimensional position of any point in space. The coordinates for these markers were digitized using the Ariel Performance Analysis System (3). For synchronization of the force and video data, the computer triggered a stroboscopic light, which was visible on the camera's field of view.

Three-dimensional marker position data were generated using the direct linear transformation method. The resulting displacement-time data of each marker were filtered using a second-order Butterworth digital filter with zero-order phase lag (33). Optimal cut-off frequencies were chosen by comparing the residuals of the difference between filtered and unfiltered signals at several cut-off frequencies (33). From the smoothed angular displacement data, the hip and knee extension/flexion position data were further analyzed (33).

Ground reaction force and EMG data were averaged over 0.60 intervals, yielding 600 data points per second. This was performed to achieve simultaneous examination and display of kinematic and kinetic data. Average EMG (mV) for the 1RM and the jumps (squats, CMJs) was calculated by full-wave rectification and averaged over eccentric and concentric phases. For each jump, GRF, EMG, and joint-displacement values were used for the analysis. For CMJ, the movement was divided into concentric and eccentric phases. These phases were determined using the changes in angular position of the knee and the contact time as proposed by Aura and Vitasalo (4) and Vitasalo et al. (29). From GRF data, the maximum VJ height was determined (29). Furthermore, power was obtained by integrating the vertical GRF for each phase of the jump. Mechanical power provides an index of the combined effect of force exerted by the subject on the ground multiplied by the velocity of the center of mass during the movement. From the kinematic raw angular displacement values, the maximum hip (MAH) and knee (MAK) values were further analyzed. Maximal angles were analyzed because they provide an indication of the maximum joint range of motion achieved during each task. Therefore, any changes in the maximum angle after training may be indicative of gross changes in movement technique. The raw EMG was full-wave rectified and averaged over eccentric and concentric phase (CMJ) or during the whole movement (SJ). The average EMG values provide an estimation of neuromuscular activation of specific muscles during each task.

Testing Procedure

Subjects underwent a standardized warm-up consisting of submaximal cycling (5 minutes), stretching exercises, and several VJs. The main protocol included performance of 3 SJs and CMJs, in a randomized order. The SJ test was performed from the seated position with the knee secured at 90° of knee

flexion (180° = full extension) while subjects kept their hands on their hips. The starting position for CMJ was with the knee at 180° (full extension). Neither verbal nor visual motivation was provided during the tests. The rest interval between trials was 3 minutes, whereas a 5-minute interval was given as the height of the jump was altered.

Statistical Analyses

An ANOVA with repeated measures was used to examine the effects of training (PRE-POST) between the 4 groups (OL, WP, PL, C) on each dependent variable. Post hoc Tukey tests were applied for comparisons between individual means, when required. The level of significance was set at $p \leq 0.05$.

RESULTS

Vertical Jump Height and Power

The results for VJ height and mean power for each testing condition are presented in Table 2. The analysis of variance (ANOVA) indicated a significant Group $\times$ Training interaction effect on SJ height ($p < 0.05$). Post hoc analysis showed that all experimental groups improved SJ height, but this was not true for the Control group. Squat jump power did not increase for all groups ($p < 0.05$). Only the WP group showed significant improvement in SJ power ($p < 0.05$) after training. For CMJ, the ANOVA indicated a significant Group $\times$ Training interaction effect on the CMJ height ($p < 0.05$). Post hoc analysis showed that all experimental groups, but not the control group, improved the CMJ height and eccentric power. Only the OL group showed a significant increase in concentric power after training ($p < 0.05$).

Joint Kinematics

The results for knee and hip angles are presented in Table 3. There was a significant Group $\times$ Training interaction effect on knee angle in SJ ($p < 0.05$). Post hoc analysis showed that the WP group increased knee joint angle after training ($p < 0.05$). For CMJ, the ANOVA indicated a significant Group $\times$ Training interaction effect on hip angle ($p < 0.05$). Post hoc analysis showed an increased hip angle after OL training ($p < 0.05$) and a decreased hip angle after WP training ($p < 0.05$).

Electromyographic Activity

The EMG results are presented in Figure 1. There was a significant Group $\times$ training interaction effect on EMG of RF in SJ ($p < 0.05$). Post hoc analysis showed that RF activity increased after OL and WP training and that GAS activity increased after PL training. For CMJ, the ANOVA indicated a significant Group $\times$ Training interaction effect on EMG of RF ($p < 0.05$). Post hoc analysis showed that RF EMG increased only for the WP group in the eccentric phase, whereas the RF EMG decreased for the PL training in the same phase. For the concentric phase of CMJ, post hoc analysis showed that the RF EMG increased only for the OL group, but it decreased after PL training ($p < 0.05$). The EMG of GAS during the eccentric phase of the CMJ increased significantly after PL training, but it decreased after WP training ($p < 0.05$). The GAS EMG during the concentric

TABLE 2. Mean $\pm$ SD group values for each experimental condition before (pre) and after (post) training.*

Test		Olympic weight lifting		Plyometrics		Combined		Control	
		Pre	Post	Pre	Post	Pre	Post	Pre	Post
SJ	MH (cm)	28.1 $\pm$ 5.7	33.8 $\pm$ 5.5†	29.7 $\pm$ 8.6	33.9 $\pm$ 7.9†	29.4 $\pm$ 7.9	33.7 $\pm$ 7.3†	29.5 $\pm$ 4.5	31.1 $\pm$ 4.6
	Power (W)	1,319 $\pm$ 43.3	1,335.8 $\pm$ 523.4	1,094.5 $\pm$ 504.9	1,194.0 $\pm$ 465.1	1,335.4 $\pm$ 632.3	1,677.7 $\pm$ 587.7	1,306.5 $\pm$ 474.7	994.6 $\pm$ 521.6
CMJ	MH (cm)	34.6 $\pm$ 7.5	39.8 $\pm$ 6.8†	31.5 $\pm$ 6.3	36.1 $\pm$ 6.4†	34.4 $\pm$ 8.3	39.6 $\pm$ 8.6†	33.3 $\pm$ 5.2	35.2 $\pm$ 5.8
	Power (w)	-514.6 $\pm$ 141.1	-807.6 $\pm$ 270.8†	-436.3 $\pm$ 110.7	-777.8 $\pm$ 242.4†	-502.2 $\pm$ 214.3	-709.1 $\pm$ 228.5†	-469 $\pm$ 160.3	-571.5 $\pm$ 189.7
CON		Power (w)	1,403.3 $\pm$ 571.5	2,093.6 $\pm$ 587.1†	1,547.7 $\pm$ 524.2	1,638.2 $\pm$ 567.2	1,714 $\pm$ 635	1,803 $\pm$ 675	1,801.6 $\pm$ 419.9

*SJ = squat jump; CMJ = countermovement jump.

†Significant difference from pre to post ($p < 0.05$).

TABLE 3. Mean $\pm$ SD group values for each experimental condition before (pre) and after (post) training.*

Test		Olympic weight lifting		Plyometrics		Combined		Control	
		Pre	Post	Pre	Post	Pre	Post	Pre	Post
SJ	MAK ($^{\circ}$)	143.3 $\pm$ 4.6	149.6 $\pm$ 16.1	143.8 $\pm$ 16.5	152.0 $\pm$ 13.3	128.1 $\pm$ 15.7	148.1 $\pm$ 16.1	148.1 $\pm$ 6.1	145.1 $\pm$ 10.4
	MAH ($^{\circ}$)	147.2 $\pm$ 7.1	150.5 $\pm$ 11.4	143.2 $\pm$ 19.1	157.3 $\pm$ 16.6	144.2 $\pm$ 11.3	146.5 $\pm$ 9.7	149.1 $\pm$ 11.8	144.2 $\pm$ 11.2
CMJ	MAK ($^{\circ}$)	126.8 $\pm$ 21.4	141.1 $\pm$ 12†	141.5 $\pm$ 17.3	129.1 $\pm$ 15†	142.3 $\pm$ 18.6	138.1 $\pm$ 13.2	134.8 $\pm$ 18.9	136.6 $\pm$ 19.1
	MAH ($^{\circ}$)	119.6 $\pm$ 9.7	147.7 $\pm$ 13†	122.7 $\pm$ 27	119.2 $\pm$ 23.2	135.3 $\pm$ 16.1	122.7 $\pm$ 18†	125.5 $\pm$ 16.1	129.5 $\pm$ 15.8

*SJ = squat jump; CMJ = countermovement jump; MAH = maximum hip value; MAH = maximum knee value.

†significant difference from pre to post ($p < 0.005$).

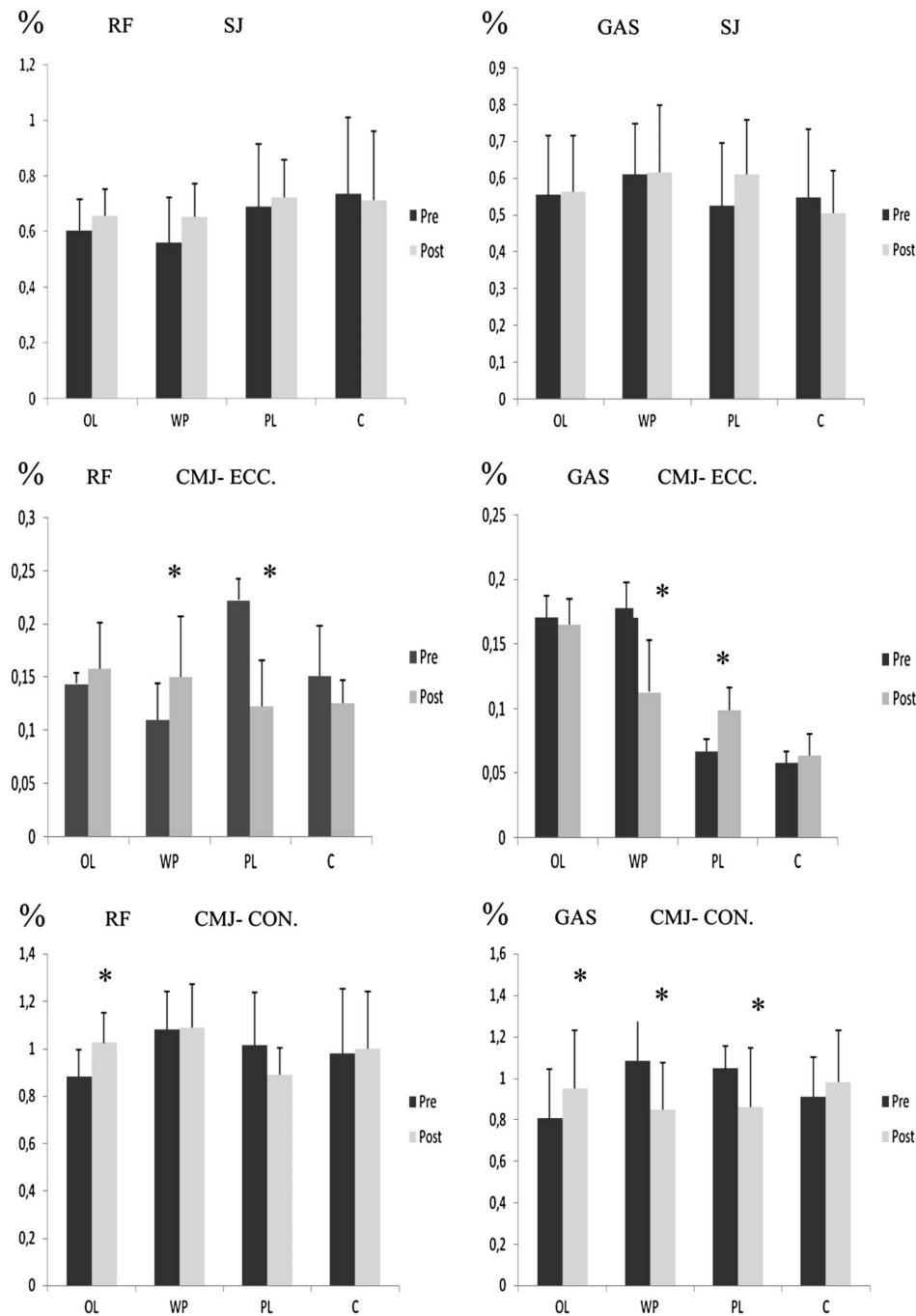


Figure 1. Training-associated changes in EMG activity of Rectus Femoris and Gastrocnemius (right and left, respectively), as assessed through squat jump (SJ), and in countermovement jump (CMJ) in the eccentric phase (ECC) and CMJ in concentric phase (CON) *Significant differences from pre to post ($p < 0.05$).

phase of the CMJ increased after OL training and decreased after WP and PL training ($p < 0.05$).

DISCUSSION

The results of this study indicated that all experimental training groups improved VJ height. However, the OL group

showed a higher increase in concentric power, whereas there were different adaptations in EMG activity among the 3 experimental groups.

In the present study, all training groups significantly improved VJ height (Table 2). For PL training, this increase can be explained via the training specificity principle, because

PL training is based on jumping exercises (24). For the OL group, improvements in jumping height can be explained because of the similarity in kinetics and kinematics between jump and OL exercises (7). For the WP group, the VJ height increase could be because of various factors, originating from each type of exercise. In other words, if both PL and OL training increased VJ height, then it is reasonable to expect a similar increase when the 2 methods are combined into a single program.

The results of this study indicated nonsignificant differences in SJ height adaptations between the 3 groups (Table 2). Particularly, all groups improved SJ, without changes in SJ power (Table 2). Furthermore, there were very few changes in either kinematic or EMG characteristics (Figure 1, Table 3) after training. These findings are consistent with some studies (22,30) and in contrast to others (14,28). This can be attributed to differences in training protocol between various studies. Despite this, our results indicate that all groups achieved a higher SJ height through improved maximum strength capacity and perhaps a better coordination.

The present results indicated that all training groups improved CMJ height (Table 2). However, there were differences in power, kinematic, and EMG adaptations between training groups. Particularly, it appears that concentric CMJ power increased for the OL group but not for the PL and WP groups (Table 2). This is in agreement with the study of Tricoli et al. (28), and it indicates that the greater skill complexity required for the OL exercises facilitates the development of a broader spectrum of mechanical parameters, which may be transferred to performance (9,13,14). It has been suggested that in OL movements, speed intensity is maximal, which may induce greater motor-unit synchronization (34). In this respect, the additional load used during OL movement execution appears to be a determining factor for power production (13). In contrast, PL jump training apparently enhances the ability to use SSC and increases the overall neural stimulation of the muscle (27). The nonsignificant change in concentric power after PL training supports previous suggestions that PL exercises do not effectively increase fundamental mechanical power, because of too small or too similar training stimuli or overreached adaptation processes (24). This does not seem to improve even when PL exercises are combined with WL training, into a single combined (WP) program.

In contrast to concentric CMJ power results, all training protocols showed a significant improvement in eccentric CMJ power (Table 2). This suggests that, despite their differences in loading method and patterns, all methods improved some or all mechanisms related to the initial (negative) phase of the CMJ. For example, Toumi et al. (27) suggested that CMJ improvements might be because of increased activation in the eccentric phase, changes in reflex activity, or increase in SSC capacity. All these

mechanisms might be active during PL (32) or OL training (28).

The results showed differences in CMJ joint angle adaptations between the 3 groups. Particularly, the OL group showed significant increases in MAH and knee joint angle (Table 3). This suggests that subjects worked through a greater range of hip and knee joint motion after OL training. Although this may not necessarily improve CMJ height, it does indicate that OL training exercises involve activation throughout the joint-displacement spectrum, which might explain the higher power observed after OL training (Table 2). If a better CMJ performance is achieved by a higher effort of the hip joint muscles (18), our results might suggest that the OL group adopted a new technique manifested by a better use of hip and knee joint muscles.

The higher eccentric power (Table 2), accompanied by a lower knee angle (Table 3) during the transition phase, indicates that the athletes who followed the PL program improved their CMJ height performance through better use of the SSC cycle, which was accompanied by smaller changes in knee and hip angles compared with the OL group. In addition, our results showed that the WP training showed a decline in maximal hip angle only (Table 3). Because OL training increased hip and knee angles and the PL group caused a decline in knee angle, the combination of these 2 programs seems to have balanced their effects on CMJ angular kinematics.

The results of this study showed that the RF and GAS EMG during the CMJ increased for the OL training group (Figure 1). It has been shown that quadriceps activation contributes 50% of the total work in a CMJ (11). Furthermore, Toumi et al. (27) reported that combined jump and weight (such as OL exercises) produce higher activation of knee extensors and improve the transfer of activation from the knee to the ankle. The above suggests that the improvements in jumping performance as a result of OL training were because of higher leg extensor activation and a change in jump strategy (as interpreted through kinematic data changes) (27).

For the PL group, the decline in RF EMG during the stretch phase and the increase of GAS during the same phase (Figure 1) provide support to the previous studies that PL training improves the ability to store and release elastic energy (6,15). Furthermore, the reduction of RF EMG combined with the increase in VJ height suggests that VJ adaptations might have been the result of improved efficiency. This increased efficiency may be indicative of training-specific effects of the PL program, in contrast to higher power and EMG adaptations observed after OL training.

The WP group showed an increase in RF EMG that is similar to OL training improvements and suggests that exercising with weights in addition to PL exercises enhances quadriceps activation. However, the reduction of GAS EMG

after WP training, coupled with the smaller kinematic changes, suggests that combined PL and OL training mainly affects knee muscle strength capacity without clear changes in the CMJ technique.

In conclusion, the mechanisms of VJ height improvement may differ among the 3 protocols because OL training has a beneficial effect on muscle strategy and maximal power performance during the CMJ, whereas PL training increases performance through improved use of the SSC and better coordination. Combined weight lifting + plyometric training compensates for adaptations from the 2 methods and enhances CMJ performance in knee extensors through increased activation of the knee extensors in the eccentric phase of the CMJ.

PRACTICAL APPLICATIONS

Improving muscle function and sport performance is of primary importance for strength and conditioning professionals. Our results suggest that all the training programs improved VJ performance. The different adaptations, however, suggest that each training program could be more useful in different periods of the training season. Olympic weight lifting exercises seem to improve VJ height via changes in power and technique, and therefore, their use in the precompetition period might be more appropriate. Plyometric exercises improve VJ height through better SSC use, and therefore, their use in the competitive period would be preferable. Combined, weight lifting + plyometric exercises are practically useful to allow an easy transition from OL to more sport-specific exercises such as the PL ones. Coaches might also prefer to use combined programs when there is a need to improve strength and power of their athletes within the competition period.

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